

1. **Integer (Z) numbers:** Together with 0:
 - a. **Natural (N) numbers:** Non-negative whole number
 - b. Negative whole numbers
2. **Real (R) numbers:**
 - a. Include:
 - i. **Rational (Q) numbers:** Can be written as the ratio of 2 integers (like $\frac{1}{2}$ or $\frac{3}{2}$). These can be written as decimals, either with finite expansion or infinite repeating pattern expansion
 - ii. **Irrational numbers:** Numbers that cannot be written as a ratio of 2 integers. These can be written as decimals with infinite non-repeating pattern expansion
 - b. Sequences such as the 1 used to approximate e will converge to an element in R
 - c. For any 2 elements on the real line, there is also a third element in between those 2, no matter how close the 2 initial elements are
3. **The Euler number (e)** is an irrational number but can be approximated by a sequence of rational numbers: $(1 + \frac{1}{n})^n$ with $n \in N$

$$\rightarrow \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$$
4. **Inequalities:**
 - a. If $a < b$ then $a + c < b + c$, for any c
 - b. If $a < b$ and $c > 0$ then $ac < bc$
 - c. If $a < b$ and $c < 0$ then $ac > bc$
5. **Sets:**
 - a. **Open Interval:** The set of numbers between 2 elements $a \in R, b \in R$, excluding a and b themselves, denoted (a,b)
 - b. **Close Interval:** The set of numbers between 2 elements $a \in R, b \in R$, including a and b themselves, denoted $[a,b]$
 - c. **Subset:** All elements of A belongs to B , indicated as $A \subseteq B$
 - d. **Proper Subset:** All elements of A belongs to B and at least 1 element of B does not belong to A , indicated as $A \subset B$
 - e. **Union:** $A \cup B = \{x \in R | x \in A \text{ or } x \in B\}$
 - f. **Intersection:** $A \cap B = \{x \in R | x \in A \text{ and } x \in B\}$
 - g. **Complement:** $A - B = A/B = \{x \in R | x \in A \text{ and } x \notin B\}$
 - h. **Empty set:** Does not contain any elements, denoted $A = \phi$
6. **Induction Principle:**
 - a. A property $P(n)$ is a statement involving elements n either belonging to N or 1 of its subsets. $P(n)$ is said to be true if it is valid for every value of n
 - b. Theorem: Support that a property $P(n)$ is true for $n = 1$. Now suppose also that if the property is assumed to be valid for a generic n , then this results in it also being valid for $n+1$. Then $P(n)$ is valid for all numbers $n \in N$

7. Functions

- a. **Domain:** Given $f: X \rightarrow Y$, indicated as $X = \text{dom}(f)$
- b. **Range:** Given $f: X \rightarrow Y$, indicated as $X = \text{ran}(f)$
- c. f is **Surjective** if $\text{ran}(f) = Y$
- d. f is **Injective** if $f(x_1) = f(x_0)$ then $x_1 = x_0$
- e. f is **Bijjective** if it is both Surjective and Injective
If a function is Bijjective, $\text{graph}(f)$ has only 1 element of $\text{dom}(f)$ mapped to it
- f. **Graph:** $\text{graph}(f) = \{(x, y) \in R^2 | x \in \text{dom}(f), y = f(x)\}$
 R^2 indicates that both x and y belong to R ($R \times R$, this is “Cartesian Product”)
- g. **Vertical line test:** A set G of points in the Cartesian plane, is a graph of function f if every vertical line intersects G at most once
- h. **Horizontal line test:** Given $f: X \rightarrow Y$, is said to be 1-to-1 or bijective if each element of the set $\text{ran}(f)$ has only 1 element of $\text{dom}(f)$ mapped to it
- i. **Composition:**
 - i. Consider $f: X \rightarrow Y$ and $g: V \rightarrow W$, assume $V \subseteq Y$ with $X, Y, W \subseteq R$
 - ii. The composition of g with f is indicated by $g \circ f$ and is the function that associates every $x \in X$ with the element $g(f(x))$ in W , if this is defined
- j. **Inverse of a function:**
 - i. If a function $f: X \rightarrow Y$ is injective:
 1. $f^{-1}(f(x)) = x$ for every $x \in X$, and
 2. $f(f^{-1}(x)) = x$ for every $y \in Y$
 - ii. **Monotonic functions** are those that are strictly increasing or decreasing, are always invertible
 - iii. The graph of a function and its inverse are symmetrical with respect to the line $y = x$